Stability aspects of single electron threshold logic based 4 bit carry look ahead adder

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***Abstract*— This paper demonstrates the detailed design of carry look ahead adder with the single electron tunneling based threshold logic. Tunneling is a mechanism in which a single electron can cross a sandwiched structure of insulating layer between two conducting materials known as tunnel junction. The threshold logic works mainly on the basis of the comparison in between the threshold and the weighted sum. Here single electron tunneling technology is used for designing threshold logic gates. The proposed circuit is simulated in SIMON environment. The simulated input and output waveforms confirm the proper functioning of the designed circuit. One of the main concerns of our design is the circuit stability aspect. The stability plots of the proposed circuit are simulated and the results are also discussed in the paper.**

Keywords—Single electron tunneling, Threshold logic ,Carry look ahead adder, SIMON, Stability Plot.

I. INTRODUCTION

The evolution of the electronic device is moving towards the improvement of electronics devices from the point of view of the ultra-thin dimension, high speed and low power consumption. The single electron tunneling(SET) devices are low dimensional devices with very low power consumption and very high speed. With these characteristics it has become one of the most promising technologies. SET[1-2] can be used in future applications as a suitable successor of the MOS technology which suffers from several issues related to scaling down. Different logic functions can be implemented by using single electron threshold logic[3-6] gates which work on the principle of comparison in between weighted sum and threshold.

Adders are one of the important basic building blocks of any complicated digital circuits. There are different types of adders among which here in this paper we have discussed about the implementation and simulation of the logical circuit of carry look ahead adder. There is a specific range of input voltage combinations of any circuit for which the complete circuit output is stable. The main focus of our work is laid upon the stability [9] aspects of the proposed circuit. The principle of operation of the threshold logic based circuit is discussed in section II. The conventional circuitry and also the implementation of 4 bit carry look ahead adder using the above mentioned approach is presented in details in the section III. The third section also consists of the simulation results of the designed circuit in the SIMON [7-8] environment. Section IV includes the stability aspects of the complete circuit.

II. SINGLE ELECTRON THRESHOLD LOGIC

The single electron threshold logic works on the principle of comparison between the weighted sum and the threshold value. Equation (1) and (2) gives the relationship of the output function with the weighted sum and threshold value T. The weighted sum in case of single electron threshold logic is calculated by summing up the products of the input voltages($x_1,x_2,x_3,\ldots x_n$) with their corresponding input capacitors($w_1,w_2,w_3,\ldots w_n$).

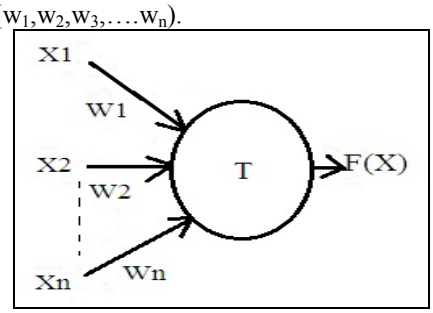


Figure 1:Basic threshold logic gate with inputs $x_1,x_2,x_3,\ldots x_n$,the input weights $w_1,w_2,w_3,\ldots w_n$,threshold T and output F(X)

$$F(X) = sgn\{M(X)\} = \begin{cases} 0 & \text{if } M(X) < 0 \\ 1 & \text{if } M(X) \geq 0 \end{cases} \quad (1)$$

$$M(X) = \sum_{i=1}^{n} w_i x_i - T \quad (2)$$

III. IMPLEMENTATION OF 4 BIT CARRY LOOK AHEAD ADDER

$$G_i = A_i.B_i \quad (3)$$

$$P_i = A_i \oplus B_i \quad (4)$$

The relation between the input signals and output signals are shown in the above two equations. Equation (3) shows the generalized expression of the output generate signal and

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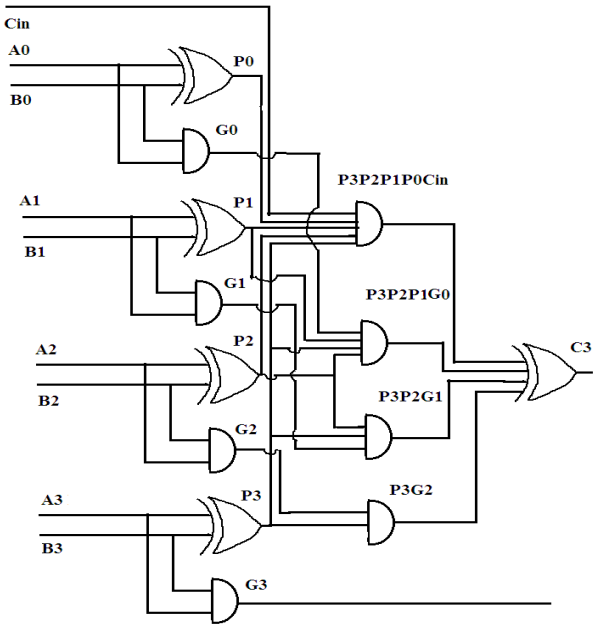


Figure2: The conventional circuit diagram of 4 bit carry look ahead adder

equation (4) shows the expression of propagate signal where A and B are the two input signals. The relationship of the different input and output signals for 4 bit carry look ahead adder is shown in the following equations. The generated carry outputs are denoted with C_0, C_1, C_2, C_3 and they are related to the different propagate as well as generate signals in Equation no.(5),(6),(7) and (8).

$$C_0 = P_0 C_{in} + G_0 \qquad (5)$$

$$C_1 = P_1 C_0 + G_1 = P_1 P_0 C_{in} + P_1 G_0 + G_1 \qquad (6)$$

$$C_2 = P_2 C_1 + G_2 = P_2 P_1 P_0 C_{in} + P_2 P_1 G_0 + P_2 G_1 + G_2 \qquad (7)$$

$$C_3 = P_3 C_2 + G_3 = P_3 P_2 P_1 P_0 C_{in} + P_3 P_2 P_1 G_0 + P_3 P_1 G_1 + P_3 G_3 \qquad (8)$$

The circuit of 4 bit carry look ahead adder mainly consists of 4 half adder for generation of sum and rest of the circuitry is used for generation of the carry. Here the complete circuit is implemented using single electron threshold logic shown in figure 3. Complete circuit design in the proposed approach requires 32 buffers, 20 two input NAND gates and 4 NOR gates. The inputs to the four half adders are A_0,B_0,A_1,B_1,A_2,B_2,A_3,B_3.For avoiding complications we have considered the input A_0,A_1,A_2,A_3 are similar andB_0,B_1,B_2,B_3 are similar in nature.

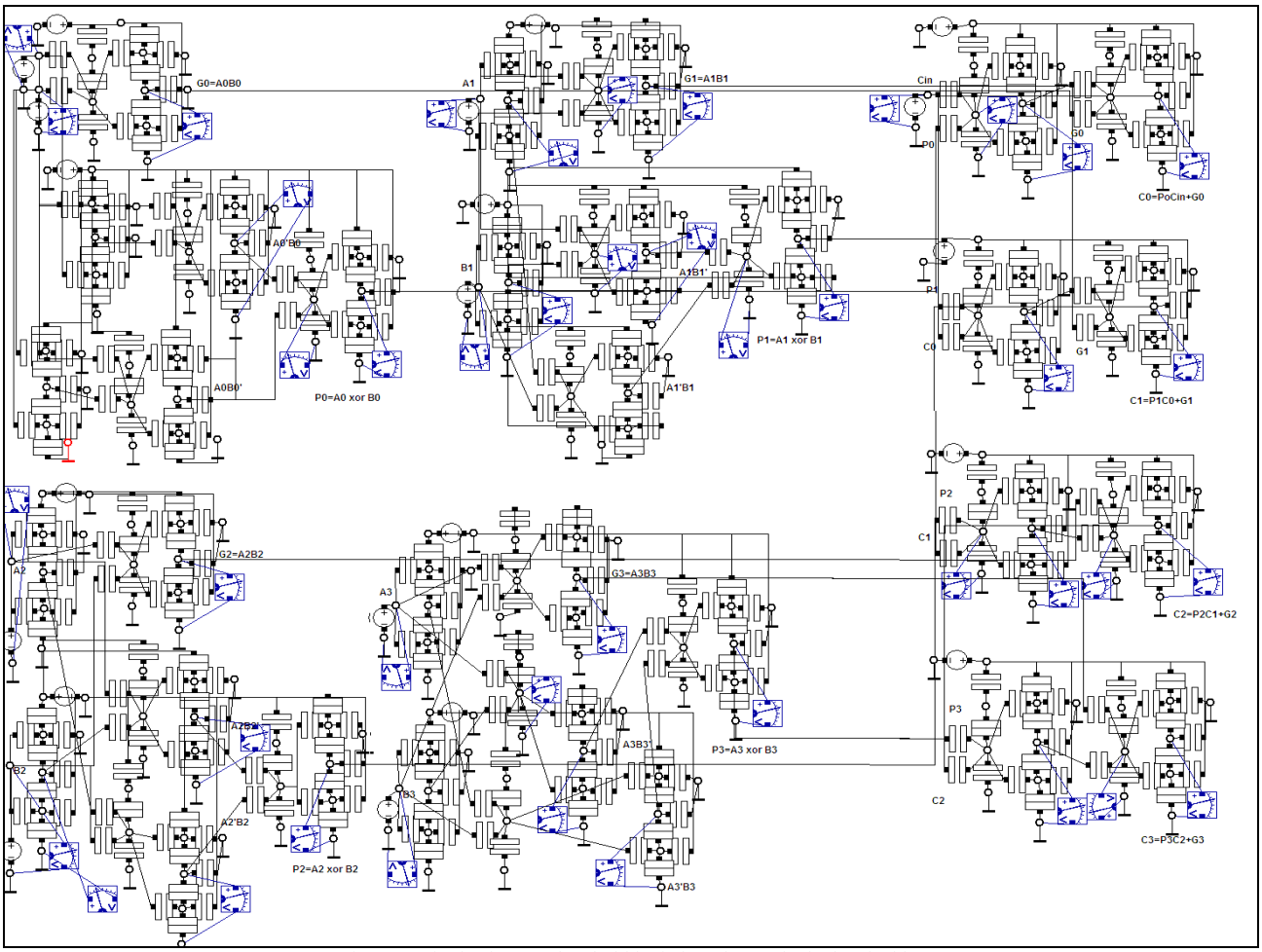


Figure 3: Single electron threshold logic based carry look ahead adder

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The figure 4 (a) and (b) shows the simulated waveform of A_0 and B_0 respectively. The output of individual half adder is represented with P_0,P_1,P_2, P_3 (propagate signals or sum terms of each half adder) and G_0,G_1,G_2,G_3(generate signals or carry signals of each half adder).Figure 4 (c) and (d) shows the simulated waveform for G_0 and P_0. As all the input A(A_0,A_1,A_2,A_3) and B (B_0,B_1,B_2,B_3)are considered to be similar ,all the generate and propagate signals are also same. According to the expressions the carry signal C_0,C_1,C_2 and C_3 are generated and the simulated waveforms are shown in Figure 4 (e), (f), (g) and (h) respectively.

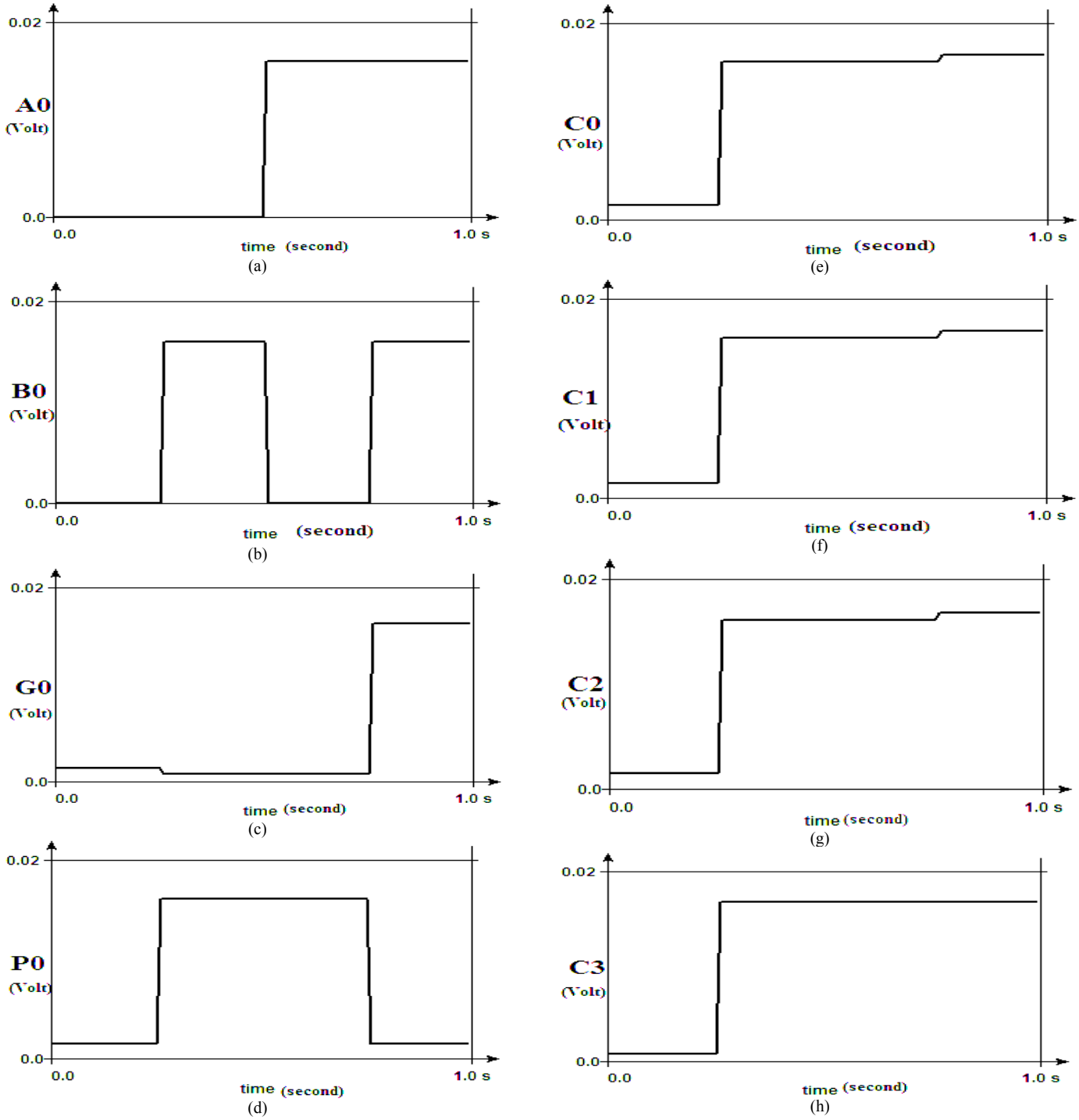


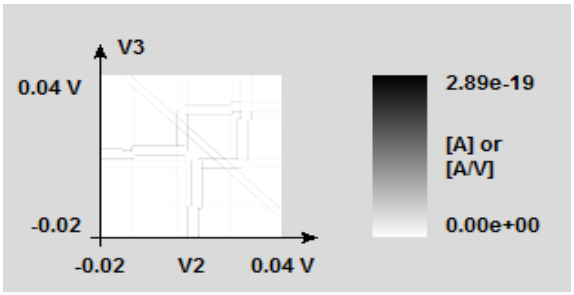
Figure 4. Simulated waveforms for (a)input A_0,(b) input B_0,(c) generate signal G_0,(d) propagate signal P_0,(e) C_0,(f) C_1,(g) C_2 and (h) C_3

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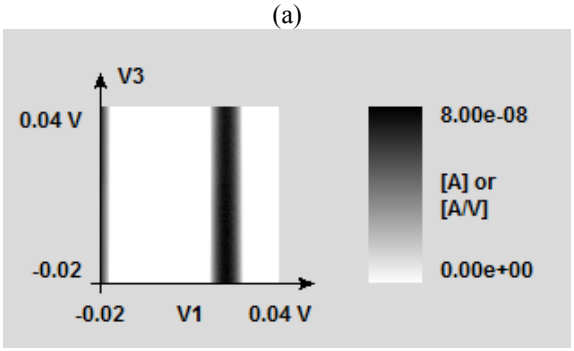
IV. Stability Analysis

The circuit stability is determined with respect to the different input values. The implemented circuit is simulated using SIMON for finding out its different stability conditions. Figure 5 (a) is the simulated stability plot for input value A_0 with

respect to input values of B_0. For both the inputs A_0 as well as B_0 the same range of voltage values are considered starting from -0.02V to 0.04V. As a result we get some points with different gray level intensities starting from white to black. The white points basically indicate the stable region with no charge fluctuation and the black or gray points indicate some amount of charge fluctuation. Depending upon the intensity of the gray points the value of charge fluctuation can be easily calculated. The first plot shown in figure 5 (a) gives the stability plot of A_0 vs B_0. Here A_0 is plotted along x axis and B_0 along the y axis of the stability plot where the simulated range of charge fluctuation we get is from 0 to 2.89e-19.similarly another stability plot can be determined by simulating the values of A_0 with respect to the power supply voltage (Vdd) shown in figure 5 (b). For this plot charge fluctuation range starts with 0 and ends at 8.00e-08. The range of voltage values are same as considered in the previous case of A_0 and B_0.



(a)



(b)

Figure 5: Stability plots of the proposed circuit for (a) A_0 with B_0 and (b) A_0 with Vdd

Another important parameter that affects the circuit stability is the temperature. The first condition that should be maintained for stable operation is given by the equation (9)

$$E_C = \frac{e^2}{2C} >> K_B T \qquad (9)$$

Where E_c is the charging energy, e is single electron charge and $K_B T$ is the thermal energy. For stable mode of operation always the thermal energy should be less than the charging energy otherwise electron tunneling will occur due to thermal fluctuations in energy.

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V. Conclusion

The circuit design presented in this paper successfully works as a 4 bit carry look ahead adder using single electron threshold logic. The complete circuit is simulated with SIMON simulator for single electron devices. All the simulated waveforms shown are verified to be logically correct. The verified results of input output waveforms satisfy all the required relationships among them. Moreover the stability aspect of the complete circuit is also discussed with the two stability plots. The stability plots show that for the selected range of input signal values the circuit is stable.

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